

Resonant versus geometric anisotropy of the Purcell enhancement near a gold nanorod: an equal-volume-sphere control and design rules for plasmon-enhanced single-photon sources

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Abstract

The decay rate of a quantum emitter near a metal nanoparticle is governed by the projected local density of optical states, and near an elongated particle this quantity depends strongly on the emitter's orientation. How much of that orientational anisotropy is a generic near-surface effect, present for any metal body, and how much is specific to the resonant longitudinal mode of the elongated particle, has not been separated quantitatively. We address this question for a dipole 5 nm from the tip of a capped gold nanorod (length 60 nm, diameter 20 nm) using three-dimensional finite-difference time-domain simulations, validated against exact Mie theory, and a set of controls: free space, a transverse dipole at the same tip, and an equal-volume gold sphere probed with radial and tangential dipoles. The longitudinal dipole reaches a Purcell factor $F_p \approx 1.5 \times 10^3$ at 610 nm and a radiative enhancement $T \approx 1.2 \times 10^2$ at 616 nm, but more than 92 % of the emitted power is absorbed in the metal. The sphere's radial-to-tangential ratio stays between 2.2 and 2.7 over the whole 500–900 nm band, consistent with the quasi-static near-surface limit of 2, and the nanorod shows the same ratio away from resonance; at the longitudinal resonance the nanorod ratio rises to 41 for F_p and to 358 for T . The orientational anisotropy of the nanorod is therefore predominantly resonant; it cannot be represented by a scalar Purcell factor and is diluted by orientation averaging. Including the intrinsic quantum yield q_0 of the emitter, we find that the optimal tip-emitter gap depends on q_0 : for poor emitters ($q_0 = 0.01$) a 5–10 nm gap raises the quantum efficiency up to eightfold, whereas for $q_0 \rightarrow 1$ the 20 nm gap performs best. An independent body-of-revolution FDTD calculation, validated against Mie theory to 4 % for the radiated power, reproduces the radiative peak of the nanorod to within 5 %.

Keywords: Purcell effect; local density of optical states; gold nanorod; plasmonic nanoantenna; quenching; single-photon source; FDTD.

1 Introduction

The spontaneous emission rate of a quantum emitter is not an intrinsic property but depends on its electromagnetic environment [1, 2]. Metallic nanoparticles supporting localized surface plasmon resonances confine light to volumes far below the diffraction limit and can enhance emission rates by several orders of magnitude [3–5]. This has made plasmonic nanoantennas a candidate platform for bright, fast single-photon sources [6–9], in which a large rate enhancement shortens the excited-state lifetime and raises the achievable repetition rate, and a short radiative lifetime relative to the dephasing time is a prerequisite for indistinguishable photons [10].

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The price of plasmonic confinement is absorption. At emitter–metal separations of a few nanometres, energy is transferred to non-radiative channels in the metal, and the fluorescence is quenched rather than enhanced [11–14]. The total decay rate and the radiative rate therefore have to be treated separately, and their ratio—the antenna quantum efficiency—depends sensitively on distance, wavelength and the intrinsic quantum yield of the emitter [15, 16].

Gold nanorods are attractive antennas because their longitudinal plasmon resonance is tunable through the aspect ratio across the visible and near infrared [17, 18], they can be synthesised with good control [19], and their tips concentrate the near field. Single-molecule experiments have demonstrated fluorescence enhancements exceeding a thousandfold near individual gold nanorods [20, 21], and theory has established that the decay-rate enhancement near the tip reaches the order of 10^3 while the efficiency falls in nanometre gaps [22].

A consequence of the nanorod’s elongated shape is a strong dependence of the response on the orientation of the emitter’s transition dipole. In the language of Eq. (1), the Green tensor at the emitter position is anisotropic, and the decay rate is the projection of that tensor on the dipole direction [2]. Yet a dipole close to any metal surface—a sphere or a plane—also shows an orientation dependence: in the quasi-static near field, energy transfer to the metal is twice as strong for a dipole perpendicular to the surface as for a parallel one, so the ratio of the two decay rates tends to two [13]. It is therefore not obvious how much of the anisotropy observed near a nanorod tip is generic and how much is caused by the longitudinal resonance that distinguishes the nanorod from a compact particle. This distinction matters in practice: emitters such as colloidal quantum dots or dye molecules are often randomly oriented, and orientation-averaged or scalar descriptions—for example a single “Purcell factor”—discard precisely the resonant part.

In this work we separate the two contributions quantitatively by introducing an equal-volume gold sphere as a control and probing both the nanorod and the sphere with two dipole orientations at an identical gap. We further include free-space normalisation, analytic Mie validation, a gap sweep and a mesh-convergence study, and we translate the results into design rules for plasmon-enhanced single-photon sources that account for the intrinsic quantum yield of the emitter and the collection numerical aperture.

2 Theory

For a point dipole $\mathbf{p} = p \hat{\mathbf{n}}$ at $\mathbf{r}_0$ in the weak-coupling regime, the total decay rate normalised to its free-space value is [2]

$$\frac{\Gamma_{\text{tot}}}{\Gamma_0} = \frac{6\pi c}{\omega} \hat{\mathbf{n}} \cdot \text{Im} \overleftrightarrow{\mathbf{G}}(\mathbf{r}_0, \mathbf{r}_0; \omega) \cdot \hat{\mathbf{n}} \equiv F_p(\omega), \quad (1)$$

where $\overleftrightarrow{\mathbf{G}}$ is the dyadic Green function. Classically, $F_p = P_{\text{tot}}/P_0$ is the power delivered by a constant-amplitude dipole relative to free space, which is how it is evaluated numerically below. The power radiated to the far field, $T = P_{\text{rad}}/P_0$, gives the radiative enhancement, and the difference $F_p - T = P_{\text{loss}}/P_0$ is absorbed in the metal. For an emitter of intrinsic quantum yield q_0 , the quantum efficiency in the presence of the antenna is [15]

$$\eta(q_0) = \frac{T}{F_p + (1 - q_0)/q_0}, \quad (2)$$

which reduces to the antenna efficiency $\eta_a = T/F_p$ for $q_0 = 1$.

The orientation average $\frac{1}{3} \text{Tr} \text{Im} \overleftrightarrow{\mathbf{G}}$ equals the projected value only for an isotropic Green tensor. For a dipole on the symmetry axis of a body of revolution, the tensor is diagonal with an axial component G_{zz} and a transverse component $G_{xx} = G_{yy}$; the anisotropy is fully characterised by the ratio $F_p^{\parallel \text{axis}}/F_p^{\perp \text{axis}}$. We define the analogous ratio for the radiative channel, $T^{\parallel}/T^{\perp}$. For a sphere, axial and transverse correspond to radial and tangential dipoles, for which exact Mie expressions exist [23–25]. The equal-volume sphere thus provides a reference in which only the

generic near-surface anisotropy is present: it has the same amount of gold, the same dipole–surface gap, but no elongated resonance.

3 Methods

3.1 Geometry and cases

The nanorod is a gold cylinder of length 40 nm and radius $R = 10$ nm terminated by two hemispherical caps (total length $L = 60$ nm, aspect ratio 3), a geometry accessible by seed-mediated growth [19]. The dipole is placed on the rod axis at a distance d from the apex ($d = 5$ nm unless stated otherwise). The equal-volume sphere has $R_{\text{sph}} = 15.874$ nm ($\frac{4}{3}\pi R_{\text{sph}}^3 = \pi R^2 \cdot 40 \text{ nm} + \frac{4}{3}\pi R^3$). The cases are: A, rod with axial (longitudinal) dipole; C, rod with transverse dipole; D and $D_{\perp}$, sphere with radial and tangential dipoles; B, free space; plus gap (3–20 nm) and mesh variations of case A. The host is air unless stated.

3.2 Gold permittivity

We use the Johnson–Christy data [26]. For the time-domain solvers it is represented by a Drude term plus two Lorentz poles fitted over 440–1000 nm (root-mean-square relative error 1.6 %, maximum 3.4 %); the fit parameters are given in the Supplementary Information. Benchmarks of the body-of-revolution solver use the same fitted permittivity in the Mie reference, so that the comparison isolates discretisation errors; comparisons with the three-dimensional solver use the tabulated data, which that solver fits with its own multi-coefficient model [27].

3.3 Numerical methods

Full three-dimensional FDTD. Cases A– $D_{\perp}$ were first computed with a commercial FDTD solver (Lumerical FDTD 2024 R1) [27, 28] with a 600 nm simulation domain, twelve PML layers, a uniform 2 nm override mesh around the particle and gap, and a 400 nm closed box of power monitors. F_p is obtained from the power injected by the dipole and T from the flux through the box, both normalised to the same dipole in a homogeneous medium.

Body-of-revolution FDTD. Because the dipole lies on the symmetry axis, the fields can be expanded as $e^{im\varphi}$ and an axial dipole couples only to $m = 0$ and a transverse dipole only to $m = \pm 1$. The three-dimensional problem then reduces to a two-dimensional one in the (r, z) half-plane without approximation. We exploit this with the open-source solver Meep [29] in cylindrical coordinates. T is the flux through a closed surface around the particle and $F_p = (P_{\text{rad}} + P_{\text{abs}})/P_0$, with the absorbed power obtained as the net flux into a closed surface that encloses only the particle and crosses the gap at mid-distance; this power-balance estimate is smoother than the local density of states sampled at the source cell, which is retained as a cross-check. Because the metal surface is staircased, F_p in a 5 nm gap converges slowly with grid spacing; we therefore compute grids of 2, 1.33 and 1 nm and extrapolate linearly to zero spacing, and validate this procedure against the exact solution for the sphere. T and the angular emission pattern are obtained in the same runs, and the angular emission pattern from a near-to-far-field transformation, from which the fraction of radiated power collected within a numerical aperture NA is obtained. To avoid normalisation errors at the edges of a broadband pulse, the 490–900 nm band is covered by three overlapping sub-band simulations of which only the central parts are retained.

3.4 Validation

For case D, the exact Mie solution [23, 24] with tabulated Johnson–Christy data gives $F_p = 772$ at 506 nm and $T = 6.58$ at 530 nm. The 2 nm-mesh three-dimensional FDTD reproduces both peak positions within 2 nm but underestimates the peak heights by 15 % and 8 % (Fig. 3). For

the body-of-revolution solver, a dielectric sphere is reproduced within 1 % at a 2 nm grid. For gold at a 5 nm gap, T is already within about 10 % of the exact result at 2 nm, whereas F_p is overestimated by a factor of about three at that grid and by 24 % at a 20 nm gap, showing that the error is confined to the near field of the staircased surface. Refining the grid to 1.33 and 1 nm reduces the median error of T to 8.4 % and 7.2 % and, after linear extrapolation to zero spacing, to 4.4 %; the same refinement leaves the median F_p error at 93 % at 1 nm (48 % extrapolated), and F_p also depends on the PML thickness at the 9 % level whereas T does not (0.1 %) (Supplementary Information). In this work the body-of-revolution solver is therefore used only for radiative quantities— T , the emission pattern and the collection efficiency—and all F_p values are taken from the three-dimensional solver, whose conformal treatment of the metal surface underestimates the exact sphere result by 15 % at a 2 nm mesh. The free-space reference yields F_p^B between 0.93 and 1.13 and T^B between 0.993 and 1.003 in the three-dimensional solver, which bounds the normalisation uncertainty of the three-dimensional results.

4 Results

4.1 Total and radiative enhancement of the axial dipole

Figure 1 shows F_p , T and η_a for cases A, C and D. For the axial dipole at the rod tip (case A), F_p peaks at 1.50×10^3 at 610 nm and T at 119 at 616 nm. The radiative peak is red shifted by 6 nm relative to the Purcell peak, because F_p contains higher-order, non-radiative multipoles that lie to the blue of the dipolar mode whereas T is dominated by the dipolar mode [11, 15]. At the Purcell peak $\eta_a = 7.6$ %, i.e. $P_{\text{loss}}/P_0 = F_p - T = 1383$ and more than 92 % of the emitted energy is absorbed. The highest efficiency, 19 %, is reached at 784 nm, far from resonance, where F_p has dropped to 69. A second, weaker F_p maximum at 512 nm ($F_p = 645$, $T \approx 1.8$) coincides with the onset of interband absorption in gold and is almost purely non-radiative. These magnitudes agree with earlier calculations for comparable rods and gaps [22]. The radiative peak is confirmed independently by the body-of-revolution solver, in which the dipole lies exactly on the axis: on grids of 2, 1.33 and 1 nm its height decreases monotonically from 149 to 134 and 125 and its position moves from 631 to 617 and 611 nm, so that the 1 nm result lies within 5 % and 5 nm of the three-dimensional value (119 at 616 nm).

4.2 Orientation anisotropy: nanorod versus equal-volume sphere

Rotating the dipole to the transverse direction at the same position (case C) reduces F_p at 610 nm by a factor of 41 and T at 616 nm by a factor of 358. The transverse dipole radiates even less than in free space ($T_{\text{max}} = 0.48$) and its own F_p maximum (239 at 508 nm) reflects absorption near the interband edge rather than a radiative resonance.

The equal-volume sphere answers whether this anisotropy is specific to the rod (Fig. 2). For the sphere, the radial-to-tangential ratio of F_p lies between 2.2 and 2.7 throughout 500–900 nm, close to the quasi-static near-surface value of two [13], and shows no resonant structure. For the nanorod, the same ratio is 2.6–2.8 below the resonance (500–530 nm)—the sphere value—but rises steeply to 41 at the longitudinal resonance. For the radiated power the geometric ratio is larger, because a tangential dipole close to a metal surface radiates poorly: the sphere gives 10–50, varying slowly with wavelength. The rod’s radiative ratio is about 4 below the resonance, i.e. not larger than the sphere’s, but reaches 358 on resonance, an order of magnitude above any value found for the sphere.

The anisotropy of the nanorod Green tensor therefore splits into two contributions: a geometric, near-surface factor of order two that any metal body produces, and a resonant factor of order 10 – 10^2 that belongs exclusively to the longitudinal ($m = 0$) mode of the elongated particle. A scalar description cannot represent the second contribution, and orientation averaging dilutes it by mixing an efficient axial channel with a strongly quenched transverse one. The comparison

Table 1: Quantum efficiency $\eta(q_0)$ at the radiative peak of the rod (axial dipole), Eq. (2), from the three-dimensional FDTD spectra. The 3-nm case used a 1-nm mesh, the others 2 nm.

gap (nm)	λ_T (nm)	$F_p(\lambda_T)$	$T_{\max}$	$q_0 = 0.01$	0.1	0.5	1
3	614	3913	252	0.063	0.064	0.064	0.064
5	616	1393	119	0.080	0.085	0.086	0.086
10	614	528	46.8	0.075	0.087	0.089	0.089
20	618	79.5	9.87	0.055	0.112	0.123	0.124

with the sphere at equal volume further shows that the enhancement near 610 nm is not a volume effect: there the rod exceeds the sphere by factors of 15.6 in F_p and 24 in T , while at their respective maxima the ratios are 2.3 and 20.

4.3 Gap dependence and the enhancement–efficiency trade-off

Increasing the gap from 5 to 10 and 20 nm (Fig. 4) leaves the radiative peak at 614–618 nm while F_p falls from 1496 to 553 and 91 and T from 119 to 47 and 9.9. The antenna efficiency at the radiative peak increases only modestly (8.6 %, 8.9 %, 12 %), whereas its spectral maximum rises from 19 % to 31 % and 74 %. The exact solution for the sphere (Fig. 4, lower row) shows the same trade-off, which is generic for nanometre gaps [30]. At 3 nm the peak F_p (7198 at 516 nm, computed with a 1 nm mesh) is entirely non-radiative ($\eta_a = 0.05$ %); at this distance surface scattering, $\gamma = \gamma_{\text{bulk}} + A v_F / R$ [31, 32], and nonlocal response [33, 34] become significant, and quasinormal-mode treatments [35, 36] are required to relate the LDOS to modal Purcell factors.

4.4 Design rule: the optimal gap depends on the emitter’s intrinsic quantum yield

For single-photon sources the relevant figure of merit depends on the operating regime. If the repetition rate is limited by the emitter, the photon rate scales with $T = F_p \eta_a$; in the common situation that a pulsed laser sets the repetition rate and the lifetime is already much shorter than the pulse period, the number of photons per pulse scales with η of Eq. (2) [6, 7]. Figure 5 and Table 1 evaluate $\eta(q_0)$ at the radiative peak for each gap. For an ideal emitter ($q_0 = 1$) the 20 nm gap is best (12.4 %) and the 3 nm gap worst (6.4 %). For a poor emitter ($q_0 = 0.01$) the order is reversed in the range 5–20 nm: the 5 nm and 10 nm gaps raise the efficiency from 1 % to about 8 %, while the 20 nm gap gives 5.5 %. The crossover follows from Eq. (2): a large F_p is needed to outcompete the intrinsic non-radiative rate $(1 - q_0)/q_0$, but beyond that point additional F_p only feeds absorption. The optimal gap therefore increases with q_0 , and the design of a nanorod antenna cannot be separated from the photophysics of the emitter.

4.5 Collection efficiency

The near-to-far-field transformation of the body-of-revolution runs gives the fraction of the radiated power emitted into a cone around the rod axis. At the radiative peak this fraction is 1.2 % for NA = 0.5, 18.0 % for NA = 0.9 and 48.3 % for the full hemisphere, identical to three decimals on all three grids. These values coincide within 7 % with those of a bare axial dipole (1.3 % and 19.4 %): the antenna multiplies the radiated power by two orders of magnitude but does not redirect it, and the emission remains a dipole pattern with a node along the rod axis. For a single-photon source this has a direct practical consequence. A rod standing along the optical axis of the collection objective wastes most of its enhancement, whereas a rod lying in the focal plane presents its dipole perpendicular to the optical axis, for which the dipolar pattern gives about 33 % at NA = 0.9—almost twice as much.

4.6 Numerical accuracy

In the three-dimensional solver, the resonance position (616(4) nm) and the efficiency at the radiative peak are stable when the override mesh is changed between 3, 2 and 1.5 nm, but the peak heights are not monotonic (Fig. S2): the peak T is 184, 119 and 165, and at 1.5 nm a spurious non-radiative feature appears. The cause is the sub-cell position of the source in a 5 nm gap together with the staircased cap: with a non-radiative rate scaling as d^{-3} [2, 13], a one-nanometre effective displacement changes the loss channel by up to a factor of two. All ratios reported above are computed with an identical mesh and source position and are insensitive to this error. The body-of-revolution solver, in which the source lies exactly on the axis, gives a monotonically converging radiative peak (149, 134, 125 for grids of 2, 1.33 and 1 nm; 7 % change between the two finest grids) and collection efficiencies that do not change at all with the grid; we therefore quote radiative quantities with an uncertainty of about 10 %. The total rate F_p remains the least certain quantity: the comparison with Mie theory bounds its error in the three-dimensional solver at about 15 % for the sphere, and the mesh study suggests up to 40 % for the sharper nanorod tip, which is why the conclusions rest on ratios, positions and the radiative channel. A boundary-element calculation, which discretises only the particle surface, would remove this limitation and is planned.

5 Discussion

The central result is that the strong orientation dependence of emission near a nanorod tip is mostly resonant. For randomly oriented emitters on the axis, the averages are $\langle F_p \rangle = (F_p^{\parallel} + 2F_p^{\perp})/3$ and likewise for T . At 610 nm this gives $\langle F_p \rangle = 523$ and $\langle T \rangle = 37.9$, three times below the axial values, while the averaged efficiency $\langle T \rangle / \langle F_p \rangle = 7.2\%$ is almost the axial one (7.6 %) because the axial channel dominates both averages. The ensemble is in fact bimodal: one third of the orientations is bright, while the transverse two thirds are quenched ($\eta_a = 0.9\%$) and radiate less than in free space ($T = 0.33$). A single enhancement factor measured on an orientation ensemble [20, 21] therefore hides a population of nearly dark emitters, which matters for single-emitter applications where each emitter’s orientation is fixed.

The equal-volume comparison also clarifies the role of shape. A compact particle of the same volume provides only the near-surface factor of order two and a weak, interband-dominated response; the elongated geometry adds a radiative resonance that is red shifted out of the interband region, where gold is less lossy. This is why the rod’s radiative enhancement exceeds the sphere’s by an order of magnitude even though the metal volume is equal.

For single-photon sources, a short lifetime (large F_p) and a high quantum efficiency are both required. Our results show that a 5–10 nm gap is preferable for emitters with low intrinsic yield, such as many colloidal quantum dots at room temperature, while near-unity emitters benefit more from larger gaps; tuning the longitudinal resonance through the aspect ratio provides an independent handle on the emission wavelength. Because the antenna leaves the dipolar emission pattern intact, the rod should lie in the focal plane of the collection optics rather than along its axis, which nearly doubles the collected fraction at NA = 0.9.

The limitations of the present study are the local-response description of gold, which is adequate at 5 nm but not in sub-nanometre gaps where tunnelling and nonlocality matter [33, 37]; the classical treatment of the emitter in the weak-coupling regime; and the restriction to on-axis emitters.

6 Conclusion

Using a capped gold nanorod and an equal-volume sphere probed with two dipole orientations at the same gap, we have separated the orientational anisotropy of the Purcell enhancement into a generic near-surface factor of about two and a resonant factor of up to 10^2 that is specific to the

longitudinal mode. More than 92% of the emitted power is absorbed at the Purcell maximum for a 5 nm gap, and the optimal gap for a single-photon source depends on the intrinsic quantum yield of the emitter. Simulation scripts, raw spectra and analysis code are openly available.

Data availability

All raw spectra, simulation scripts (Lumerical and Meep) and analysis code are available at <https://github.com/aminsadidi/maghale> (a Zenodo DOI will be assigned at publication).

Author contributions

[to be completed by the authors.]

Conflict of interest

The authors declare no competing interests.

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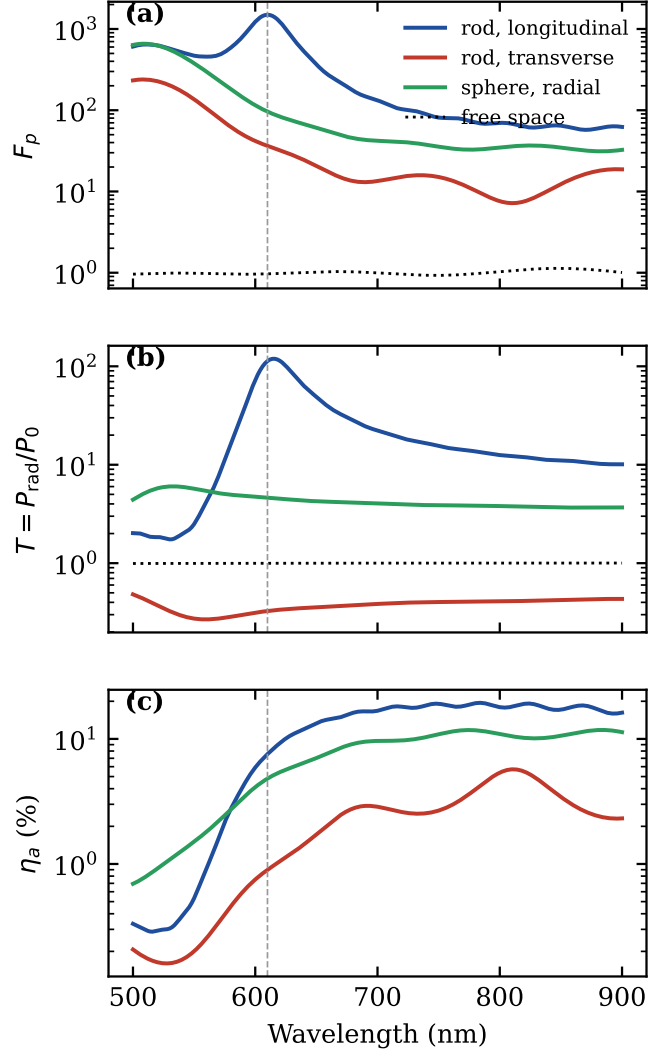


Figure 1: (a) Purcell factor, (b) radiative enhancement and (c) antenna efficiency for the rod with axial (A) and transverse (C) dipole and the equal-volume sphere with radial dipole (D), all at a 5 nm gap; dotted: free-space reference B. Dashed line: 610 nm.

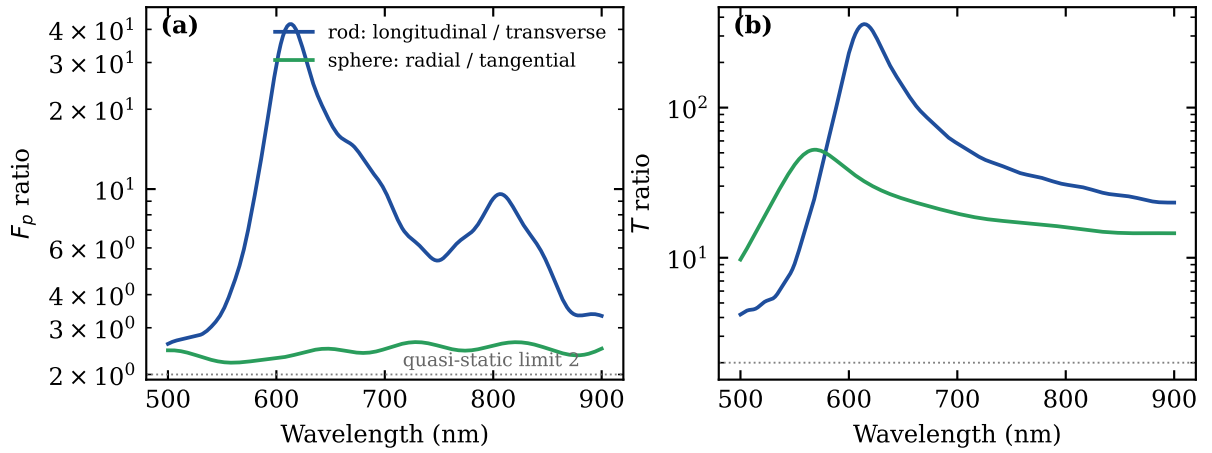


Figure 2: Orientation ratios of (a) F_p and (b) T for the rod (axial/transverse) and the equal-volume sphere (radial/tangential) at a 5 nm gap. Dotted: quasi-static near-surface limit of 2.

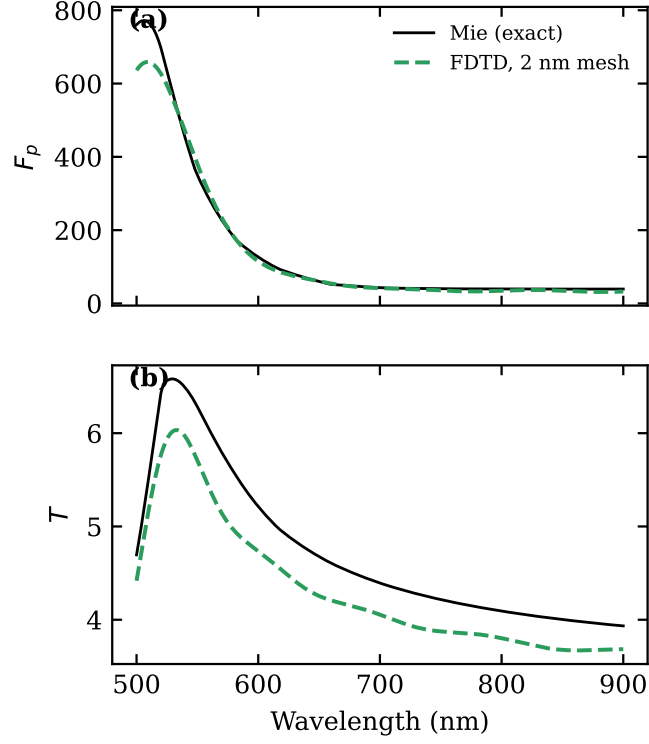


Figure 3: Validation against exact Mie theory for the equal-volume sphere with radial dipole at 5 nm: (a) F_p , (b) T .

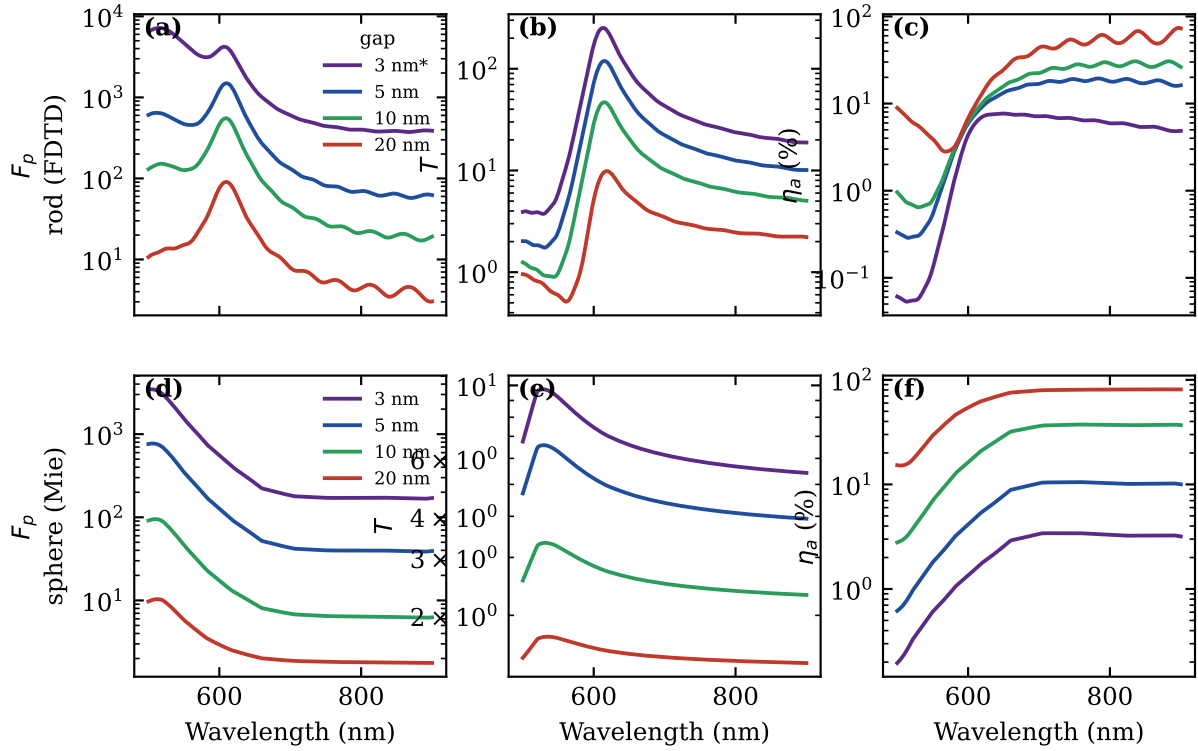


Figure 4: Gap dependence of F_p , T and η_a for the rod with axial dipole (top, FDTD; *1-nm mesh) and the equal-volume sphere with radial dipole (bottom, exact Mie).

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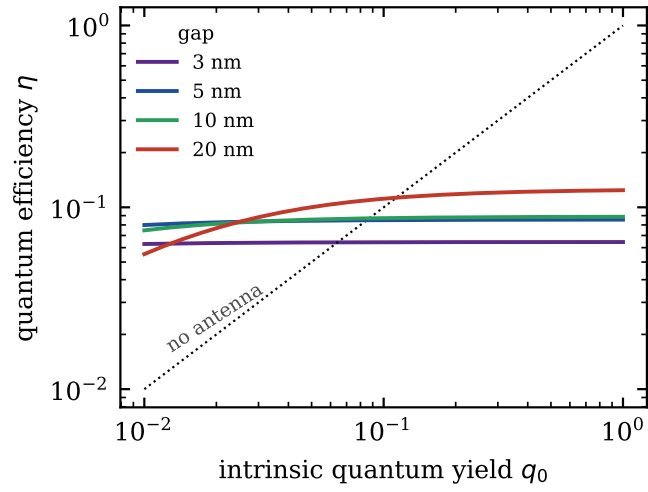


Figure 5: Quantum efficiency at the radiative peak versus intrinsic quantum yield, Eq. (2), for four gaps. Dotted: emitter without antenna ($\eta = q_0$).